

Namespace ASE_Project_Demo

Classes

[Form1](#)

Class Form1

Namespace: [ASE Project Demo](#)

Assembly: ASE Project Demo.dll

```
public class Form1 : Form, IDropTarget, ISynchronizeInvoke, IWin32Window,
    IBindableComponent, IComponent, IDisposable, IContainerControl
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← [ScrollableControl](#) ← [ContainerControl](#) ← [Form](#) ← Form1

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#), [IContainerControl](#)

Inherited Members

[Form.SetVisibleCore\(bool\)](#), [Form.Activate\(\)](#), [Form.ActivateMdiChild\(Form\)](#), [Form.AddOwnedForm\(Form\)](#), [Form.AdjustFormScrollbars\(bool\)](#), [Form.Close\(\)](#), [Form.CreateAccessibilityInstance\(\)](#), [Form.CreateControlsInstance\(\)](#), [Form.CreateHandle\(\)](#), [Form.DefWndProc\(ref Message\)](#), [Form.ProcessMnemonic\(char\)](#), [Form.CenterToParent\(\)](#), [Form.CenterToScreen\(\)](#), [Form.LayoutMdi\(MdiLayout\)](#), [Form.OnActivated\(EventArgs\)](#), [Form.OnBackgroundImageChanged\(EventArgs\)](#), [Form.OnBackgroundImageLayoutChanged\(EventArgs\)](#), [Form.OnClosing\(CancelEventArgs\)](#), [Form.OnClosed\(EventArgs\)](#), [Form.OnFormClosing\(FormClosingEventArgs\)](#), [Form.OnFormClosed\(FormClosedEventArgs\)](#), [Form.OnCreateControl\(\)](#), [Form.OnDeactivate\(EventArgs\)](#), [Form.OnEnabledChanged\(EventArgs\)](#), [Form.OnEnter\(EventArgs\)](#), [Form.OnFontChanged\(EventArgs\)](#), [Form.OnHandleCreated\(EventArgs\)](#), [Form.OnHandleDestroyed\(EventArgs\)](#), [Form.OnHelpButtonClicked\(CancelEventArgs\)](#), [Form.OnLayout\(LayoutEventArgs\)](#), [Form.OnLoad\(EventArgs\)](#), [Form.OnMaximizedBoundsChanged\(EventArgs\)](#), [Form.OnMaximumSizeChanged\(EventArgs\)](#), [Form.OnMinimumSizeChanged\(EventArgs\)](#), [Form.OnInputLanguageChanged\(InputLanguageChangedEventArgs\)](#), [Form.OnInputLanguageChanging\(InputLanguageChangingEventArgs\)](#), [Form.OnVisibleChanged\(EventArgs\)](#), [Form.OnMdiChildActivate\(EventArgs\)](#), [Form.OnMenuStart\(EventArgs\)](#), [Form.OnMenuComplete\(EventArgs\)](#), [Form.OnPaint\(PaintEventArgs\)](#), [Form.OnResize\(EventArgs\)](#), [Form.OnDpiChanged\(DpiChangedEventArgs\)](#), [Form.OnGetDpiScaledSize\(int, int, ref Size\)](#), [Form.OnRightToLeftLayoutChanged\(EventArgs\)](#), [Form.OnShown\(EventArgs\)](#),

[Form.OnTextChanged\(EventArgs\)](#), [Form.ProcessCmdKey\(ref Message, Keys\)](#), [Form.ProcessDialogKey\(Keys\)](#), [Form.ProcessDialogChar\(char\)](#), [Form.ProcessKeyPreview\(ref Message\)](#), [Form.ProcessTabKey\(bool\)](#), [Form.RemoveOwnedForm\(Form\)](#), [Form.Select\(bool, bool\)](#), [Form.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#), [Form.ScaleControl\(SizeF, BoundsSpecified\)](#), [Form.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#), [Form.SetClientSizeCore\(int, int\)](#), [Form.SetDesktopBounds\(int, int, int, int\)](#), [Form.SetDesktopLocation\(int, int\)](#), [Form.Show\(IWin32Window\)](#), [Form.ShowDialog\(\)](#), [Form.ShowDialog\(IWin32Window\)](#), [Form.ToString\(\)](#), [Form.UpdateDefaultButton\(\)](#), [Form.OnResizeBegin\(EventArgs\)](#), [Form.OnResizeEnd\(EventArgs\)](#), [Form.OnStyleChanged\(EventArgs\)](#), [Form.ValidateChildren\(\)](#), [Form.ValidateChildren\(ValidationConstraints\)](#), [Form.WndProc\(ref Message\)](#), [Form.AcceptButton](#), [Form.ActiveForm](#), [Form.ActiveMdiChild](#), [Form.AllowTransparency](#), [Form.AutoScroll](#), [Form.AutoSize](#), [Form.AutoSizeMode](#), [Form.AutoValidate](#), [Form.BackColor](#), [Form.FormBorderStyle](#), [Form.CancelButton](#), [Form.ClientSize](#), [Form.ControlBox](#), [Form.CreateParams](#), [Form.DefaultImeMode](#), [Form.DefaultSize](#), [Form.DesktopBounds](#), [Form.DesktopLocation](#), [Form.DialogResult](#), [Form.HelpButton](#), [Form.Icon](#), [Form.IsMdiChild](#), [Form.IsMdiContainer](#), [Form.IsRestrictedWindow](#), [Form.KeyPreview](#), [Form.Location](#), [Form.MaximizedBounds](#), [Form.MaximumSize](#), [Form.MainMenuStrip](#), [Form.MinimumSize](#), [Form.MaximizeBox](#), [Form.MdiChildren](#), [Form.MdiChildrenMinimizedAnchorBottom](#), [Form.MdiParent](#), [Form.MinimizeBox](#), [Form.Modal](#), [Form.Opacity](#), [Form.OwnedForms](#), [Form.Owner](#), [Form.RestoreBounds](#), [Form.RightToLeftLayout](#), [Form.ShowInTaskbar](#), [Form.ShowIcon](#), [Form.ShowWithoutActivation](#), [Form.Size](#), [Form.SizeGripStyle](#), [Form.StartPosition](#), [Form.Text](#), [Form.TopLevel](#), [Form.TopMost](#), [Form.TransparencyKey](#), [Form.WindowState](#), [Form.AutoSizeChanged](#), [Form.AutoValidateChanged](#), [Form.HelpButtonClicked](#), [Form.MaximizedBoundsChanged](#), [Form.MaximumSizeChanged](#), [Form.MinimumSizeChanged](#), [Form.Activated](#), [Form.Deactivate](#), [Form.FormClosing](#), [Form.FormClosed](#), [Form.Load](#), [Form.MdiChildActivate](#), [Form.MenuComplete](#), [Form.MenuStart](#), [Form.InputLanguageChanged](#), [Form.InputLanguageChanging](#), [Form.RightToLeftLayoutChanged](#), [Form.Shown](#), [Form.DpiChanged](#), [Form.ResizeBegin](#), [Form.ResizeEnd](#), [ContainerControl.OnAutoValidateChanged\(EventArgs\)](#), [ContainerControl.OnParentChanged\(EventArgs\)](#), [ContainerControl.PerformAutoScale\(\)](#), [ContainerControl.RescaleConstantsForDpi\(int, int\)](#), [ContainerControl.Validate\(\)](#), [ContainerControl.Validate\(bool\)](#), [ContainerControl.AutoScaleDimensions](#), [ContainerControl.AutoScaleFactor](#), [ContainerControl.AutoScaleMode](#), [ContainerControl.BindingContext](#), [ContainerControl.CanEnableIme](#), [ContainerControl.ActiveControl](#), [ContainerControl.CurrentAutoScaleDimensions](#), [ContainerControl.ParentForm](#), [ScrollableControl.ScrollStateAutoScrolling](#), [ScrollableControl.ScrollStateHScrollVisible](#), [ScrollableControl.ScrollStateVScrollVisible](#), [ScrollableControl.ScrollStateUserHasScrolled](#), [ScrollableControl.ScrollStateFullDrag](#),

[ScrollableControl.GetScrollState\(int\)](#), [ScrollableControl.OnMouseWheel\(MouseEventArgs\)](#),
[ScrollableControl.OnRightToLeftChanged\(EventArgs\)](#),
[ScrollableControl.OnPaintBackground\(PaintEventArgs\)](#),
[ScrollableControl.OnPaddingChanged\(EventArgs\)](#), [ScrollableControl.SetDisplayRectLocation\(int, int\)](#),
[ScrollableControl.ScrollControlIntoView\(Control\)](#), [ScrollableControl.ScrollToControl\(Control\)](#),
[ScrollableControl.OnScroll\(ScrollEventArgs\)](#), [ScrollableControl.SetAutoScrollMargin\(int, int\)](#),
[ScrollableControl.SetScrollState\(int, bool\)](#), [ScrollableControl.AutoScrollMargin](#),
[ScrollableControl.AutoScrollPosition](#), [ScrollableControl.AutoScrollMinSize](#),
[ScrollableControl.DisplayRectangle](#), [ScrollableControl.HScroll](#), [ScrollableControl.HorizontalScroll](#),
[ScrollableControl.VScroll](#), [ScrollableControl.VerticalScroll](#), [ScrollableControl.Scroll](#),
[Control.GetAccessibilityObjectById\(int\)](#), [Control.SetAutoSizeMode\(AutoSizeMode\)](#),
[Control.GetAutoSizeMode\(\)](#), [Control.GetPreferredSize\(Size\)](#),
[Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#),
[Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#), [Control.BeginInvoke\(Delegate\)](#),
[Control.BeginInvoke\(Action\)](#), [Control.BeginInvoke\(Delegate, params object\[\]\)](#),
[Control.BringToFront\(\)](#), [Control.Contains\(Control\)](#), [Control.CreateGraphics\(\)](#),
[Control.CreateControl\(\)](#), [Control.DestroyHandle\(\)](#), [Control.DoDragDrop\(object, DragDropEffects\)](#),
[Control.DrawToBitmap\(Bitmap, Rectangle\)](#), [Control.EndInvoke\(IAsyncResult\)](#), [Control.FindForm\(\)](#),
[Control.GetTopLevel\(\)](#), [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#),
[Control.RaiseMouseEvent\(object, MouseEventArgs\)](#), [Control.Focus\(\)](#),
[Control.FromChildHandle\(IntPtr\)](#), [Control.FromHandle\(IntPtr\)](#),
[Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#), [Control.GetChildAtPoint\(Point\)](#),
[Control.GetContainerControl\(\)](#), [Control.GetNextControl\(Control, bool\)](#),
[Control.GetStyle\(ControlStyles\)](#), [Control.Hide\(\)](#), [Control.InitLayout\(\)](#), [Control.Invalidate\(Region\)](#),
[Control.Invalidate\(Region, bool\)](#), [Control.Invalidate\(\)](#), [Control.Invalidate\(bool\)](#),
[Control.Invalidate\(Rectangle\)](#), [Control.Invalidate\(Rectangle, bool\)](#), [Control.Invoke\(Action\)](#),
[Control.Invoke\(Delegate\)](#), [Control.Invoke\(Delegate, params object\[\]\)](#),
[Control.Invoke<T>\(Func<T>\)](#), [Control.InvokePaint\(Control, PaintEventArgs\)](#),
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#), [Control.IsKeyLocked\(Keys\)](#),
[Control.IsInputChar\(char\)](#), [Control.IsInputKey\(Keys\)](#), [Control.IsMnemonic\(char, string\)](#),
[Control.LogicalToDeviceUnits\(int\)](#), [Control.LogicalToDeviceUnits\(Size\)](#),
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#), [Control.NotifyInvalidate\(Rectangle\)](#),
[Control.InvokeOnClick\(Control, EventArgs\)](#), [Control.OnAutoSizeChanged\(EventArgs\)](#),
[Control.OnBackColorChanged\(EventArgs\)](#), [Control.OnBindingContextChanged\(EventArgs\)](#),
[Control.OnCausesValidationChanged\(EventArgs\)](#), [Control.OnContextMenuStripChanged\(EventArgs\)](#),
[Control.OnCursorChanged\(EventArgs\)](#), [Control.OnDockChanged\(EventArgs\)](#),
[Control.OnForeColorChanged\(EventArgs\)](#), [Control.OnNotifyMessage\(Message\)](#),
[Control.OnParentBackColorChanged\(EventArgs\)](#),
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#),
[Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#),

[Control.OnParentEnabledChanged\(EventArgs\)](#), [Control.OnParentFontChanged\(EventArgs\)](#),
[Control.OnParentForeColorChanged\(EventArgs\)](#), [Control.OnParentRightToLeftChanged\(EventArgs\)](#),
[Control.OnParentVisibleChanged\(EventArgs\)](#), [Control.OnPrint\(PaintEventArgs\)](#),
[Control.OnTabIndexChanged\(EventArgs\)](#), [Control.OnTabStopChanged\(EventArgs\)](#),
[Control.OnClick\(EventArgs\)](#), [Control.OnClientSizeChanged\(EventArgs\)](#),
[Control.OnControlAdded\(ControlEventArgs\)](#), [Control.OnControlRemoved\(ControlEventArgs\)](#),
[Control.OnLocationChanged\(EventArgs\)](#), [Control.OnDoubleClick\(EventArgs\)](#),
[Control.OnDragEnter\(DragEventArgs\)](#), [Control.OnDragOver\(DragEventArgs\)](#),
[Control.OnDragLeave\(EventArgs\)](#), [Control.OnDragDrop\(DragEventArgs\)](#),
[Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#), [Control.InvokeGotFocus\(Control, EventArgs\)](#),
[Control.OnGotFocus\(EventArgs\)](#), [Control.OnHelpRequested\(HelpEventArgs\)](#),
[Control.OnInvalidated\(InvalidateEventArgs\)](#), [Control.OnKeyDown\(KeyEventArgs\)](#),
[Control.OnKeyPress\(KeyPressEventArgs\)](#), [Control.OnKeyUp\(KeyEventArgs\)](#),
[Control.OnLeave\(EventArgs\)](#), [Control.InvokeLostFocus\(Control, EventArgs\)](#),
[Control.OnLostFocus\(EventArgs\)](#), [Control.OnMarginChanged\(EventArgs\)](#),
[Control.OnMouseDoubleClick\(MouseEventArgs\)](#), [Control.OnMouseClick\(MouseEventArgs\)](#),
[Control.OnMouseCaptureChanged\(EventArgs\)](#), [Control.OnMouseDown\(MouseEventArgs\)](#),
[Control.OnMouseEnter\(EventArgs\)](#), [Control.OnMouseLeave\(EventArgs\)](#),
[Control.OnDpiChangedBeforeParent\(EventArgs\)](#), [Control.OnDpiChangedAfterParent\(EventArgs\)](#),
[Control.OnMouseHover\(EventArgs\)](#), [Control.OnMouseMove\(MouseEventArgs\)](#),
[Control.OnMouseUp\(MouseEventArgs\)](#), [Control.OnMove\(EventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#),
[Control.OnSizeChanged\(EventArgs\)](#), [Control.OnChangeUICues\(UICuesEventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#),
[Control.PointToClient\(Point\)](#), [Control.PointToScreen\(Point\)](#),
[Control.PreProcessMessage\(ref Message\)](#), [Control.PreProcessControlMessage\(ref Message\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(IntPtr, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.Select\(\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),

[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),
[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.OnImeModeChanged\(EventArgs\)](#),
[Control.AccessibilityObject](#), [Control.AccessibleDefaultActionDescription](#),
[Control.AccessibleDescription](#), [Control.AccessibleName](#), [Control.AccessibleRole](#),
[Control.AllowDrop](#), [Control.Anchor](#), [Control.AutoScrollOffset](#), [Control.LayoutEngine](#),
[Control.BackgroundImage](#), [Control.BackgroundImageLayout](#), [Control.Bottom](#), [Control.Bounds](#),
[Control.CanFocus](#), [Control.CanRaiseEvents](#), [Control.CanSelect](#), [Control.Capture](#),
[Control.CausesValidation](#), [Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#),
[Control.CompanyName](#), [Control.ContainsFocus](#), [Control.ContextMenuStrip](#), [Control.Controls](#),
[Control.Created](#), [Control.Cursor](#), [Control.DataBindings](#), [Control.DefaultBackColor](#),
[Control.DefaultCursor](#), [Control.DefaultFont](#), [Control.DefaultForeColor](#), [Control.DefaultMargin](#),
[Control.DefaultMaximumSize](#), [Control.DefaultMinimumSize](#), [Control.DefaultPadding](#),
[Control.DeviceDpi](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#),
[Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#),
[Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#),
[Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#),
[Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Margin](#),
[Control.ModifierKeys](#), [Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#),
[Control.Parent](#), [Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#),
[Control.Region](#), [Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#),
[Control.RightToLeft](#), [Control.ScaleChildren](#), [Control.Site](#), [Control.TabIndex](#), [Control.TabStop](#),
[Control.Tag](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#),
[Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#),
[Control.PreferredSize](#), [Control.Padding](#), [Control.ImeMode](#), [Control.ImeModeBase](#),
[Control.PropagatingImeMode](#), [Control.BackColorChanged](#), [Control.BackgroundImageChanged](#),
[Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#),
[Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#),
[Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#),
[Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#),
[Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#),
[Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#),
[Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#),
[Control.ControlAdded](#), [Control.ControlRemoved](#), [Control.DragDrop](#), [Control.DragEnter](#),
[Control.DragOver](#), [Control.DragLeave](#), [Control.GiveFeedback](#), [Control.HandleCreated](#),
[Control.HandleDestroyed](#), [Control.HelpRequested](#), [Control.Invalidated](#),
[Control.PaddingChanged](#), [Control.Paint](#), [Control.QueryContinueDrag](#),
[Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#), [Control.Enter](#), [Control.GotFocus](#),

[Control.KeyDown](#) , [Control.KeyPress](#) , [Control.KeyUp](#) , [Control.Layout](#) , [Control.Leave](#) , [Control.LostFocus](#) , [Control.MouseClick](#) , [Control.MouseDoubleClick](#) , [Control.MouseCaptureChanged](#) , [Control.MouseDown](#) , [Control.MouseEnter](#) , [Control.MouseLeave](#) , [Control.DpiChangedBeforeParent](#) , [Control.DpiChangedAfterParent](#) , [Control.MouseHover](#) , [Control.MouseMove](#) , [Control.MouseUp](#) , [Control.MouseWheel](#) , [Control.Move](#) , [Control.PreviewKeyDown](#) , [Control.Resize](#) , [Control.ChangeUICues](#) , [Control.StyleChanged](#) , [Control.SystemColorsChanged](#) , [Control.Validating](#) , [Control.Validated](#) , [Control.ParentChanged](#) , [Control.ImeModeChanged](#) , [Component.Dispose\(\)](#) , [Component.GetService\(Type\)](#) , [Component.Container](#) , [Component.DesignMode](#) , [Component.Events](#) , [Component.Disposed](#) , [MarshalByRefObject.GetLifetimeService\(\)](#) , [MarshalByRefObject.InitializeLifetimeService\(\)](#) , [MarshalByRefObject.MemberwiseClone\(bool\)](#) , [object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#)

Constructors

Form1()

```
public Form1()
```

Methods

Dispose(bool)

Clean up any resources being used.

```
protected override void Dispose(bool disposing)
```

Parameters

disposing [bool](#)

true if managed resources should be disposed; otherwise, false.

InitialiseShapes()

Create unrestricted drawing

```
public void InitialiseShapes()
```

button1_Click(object, EventArgs)

Action to be taken when the button is clicked. This will parse the program and run it.

```
public void button1_Click(object sender, EventArgs e)
```

Parameters

sender [object](#)

e [EventArgs](#)

pictureBox1_Paint(object, PaintEventArgs)

To display the canvas output on the form

```
public void pictureBox1_Paint(object sender, PaintEventArgs e)
```

Parameters

sender [object](#)

e [PaintEventArgs](#)

Namespace ASE_Project_Demo.Classes

Classes

[AppCanvas](#)

Class used to draw shapes on the canvas

[AppCommandFactory](#)

Overrides the MakeCommand method to return a new object

[AppHex](#)

This class is used to draw a hexagon on the canvas

Class AppCanvas

Namespace: [ASE Project Demo.Classes](#)

Assembly: ASE Project Demo.dll

Class used to draw shapes on the canvas

```
public class AppCanvas : ICanvas
```

Inheritance

[object](#) ← AppCanvas

Implements

ICanvas

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

AppCanvas()

```
public AppCanvas()
```

Properties

PenColour

```
public object PenColour { get; set; }
```

Property Value

[object](#)

Xpos

```
public int Xpos { get; set; }
```

Property Value

[int](#)

Ypos

```
public int Ypos { get; set; }
```

Property Value

[int](#)

Methods

Circle(int, bool)

Draws a circle with the specified radius

```
public void Circle(int radius, bool filled)
```

Parameters

radius [int](#)

filled [bool](#)

Exceptions

CanvasException

Clear()

Clears the canvas

```
public void Clear()
```

DrawTo(int, int)

This method draws a line from the current position of the pen to the specified coordinates

```
public void DrawTo(int toX, int toY)
```

Parameters

toX [int](#)

toY [int](#)

Exceptions

CanvasException

MoveTo(int, int)

Pass in the x and y coordinates to move the pen to that position

```
public void MoveTo(int x, int y)
```

Parameters

x [int](#)

y [int](#)

Exceptions

CanvasException

Rect(int, int, bool)

Creates a rectangle with the specified width and height

```
public void Rect(int width, int height, bool filled)
```

Parameters

width [int](#)

height [int](#)

filled [bool](#)

Exceptions

CanvasException

Reset()

Resets the pen position to 0,0

```
public void Reset()
```

Set(int, int)

Sets the canvas size to the specified X and Y

```
public void Set(int XSize, int YSize)
```

Parameters

XSize [int](#)

YSize [int](#)

SetColour(int, int, int)

Set the colour of the pen

```
public void SetColour(int red, int green, int blue)
```

Parameters

red [int](#)

green [int](#)

blue [int](#)

Exceptions

CanvasException

Tri(int, int)

Creates a triangle with the specified width and height

```
public void Tri(int width, int height)
```

Parameters

width [int](#)

height [int](#)

Exceptions

CanvasException

WriteText(string)

Writes text to the canvas

```
public void WriteText(string text)
```

Parameters

text [string](#)

getBitmap()

Gets the bitmap object

```
public object getBitmap()
```

Returns

[object](#)

Class AppCommandFactory

Namespace: [ASE Project Demo.Classes](#)

Assembly: ASE Project Demo.dll

Overrides the MakeCommand method to return a new object

```
public class AppCommandFactory : CommandFactory, ICommandFactory
```

Inheritance

[object](#)  ← CommandFactory ← AppCommandFactory

Implements

ICommandFactory

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

AppCommandFactory()

```
public AppCommandFactory()
```

Methods

MakeCommand(string)

Overrides the MakeCommand method to return a new object

```
public override ICommand MakeCommand(string commandType)
```

Parameters

commandType [string](#) 

Returns

ICommand

Class AppHex

Namespace: [ASE Project Demo.Classes](#)

Assembly: ASE Project Demo.dll

This class is used to draw a hexagon on the canvas

```
public class AppHex : CommandTwoParameters, ICommand
```

Inheritance

[object](#)  ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← AppHex

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed ,
CommandOneParameter.param1 , CommandOneParameter.param1unprocessed ,
CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas ,
Command.program , Command.parameterList , Command.parameters , Command.paramsint ,
[Command.Set\(StoredProgram, string\)](#)  , [Command.ProcessParameters\(string\)](#)  , Command.ToString() ,
Command.Program , Command.Name , Command.ParameterList , Command.Parameters ,
Command.Paramsint , [object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#) 

Constructors

AppHex()

```
public AppHex()
```

AppHex(Canvas, int, bool)

Draw a hexagon with the specified size

```
public AppHex(Canvas c, int radius, bool filled)
```

Parameters

c Canvas

radius [int](#)

filled [bool](#)

Exceptions

CanvasException

Methods

CheckParameters(string[])

```
public override void CheckParameters(string[] parameter)
```

Parameters

parameter [string](#)[]

Compile()

```
public override void Compile()
```

Execute()

```
public override void Execute()
```

Hex(int, bool)

```
public void Hex(int radius, bool filled)
```

Parameters

radius [int](#)

filled [bool](#)

Namespace UnitTestASE

Classes

[UnitTest1](#)

Class UnitTest1

Namespace: [UnitTestASE](#)

Assembly: UnitTestASE.dll

```
[TestClass]  
public class UnitTest1
```

Inheritance

[object](#)  ← UnitTest1

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Methods

DrawTo_ValidCoordinates_UpdatesPenPosition()

Verifies that **DrawTo** updates the pen position when given valid coordinates. Expects a **CanvasException** if the position is outside the canvas bounds.

```
[TestMethod]  
[ExpectedException(typeof(CanvasException))]  
public void DrawTo_ValidCoordinates_UpdatesPenPosition()
```

Exceptions

CanvasException

Thrown when coordinates are out of bounds or when the actual position does not match the expected values.

MoveTo_ValidCoordinates_UpdatesPenPosition()

Tests that **MoveTo** correctly updates the pen position with valid coordinates. Expects a **CanvasException** if coordinates exceed the defined canvas boundaries.

```
[TestMethod]
[ExpectedException(typeof(CanvasException))]
public void MoveTo_ValidCoordinates_UpdatesPenPosition()
```

Exceptions

CanvasException

Thrown when the coordinates are out of bounds or if the position does not match the expected coordinates.

Multi_ValidCoordinates_UpdatesPenPosition()

Tests both **MoveTo** and **DrawTo** with valid coordinates to confirm that each method updates the pen position correctly on the canvas. Throws a **CanvasException** if coordinates exceed canvas boundaries.

```
[TestMethod]
public void Multi_ValidCoordinates_UpdatesPenPosition()
```

Exceptions

CanvasException

Thrown when coordinates are out of bounds or actual position does not match expected values after each operation.

Setup()

```
[TestInitialize]
public void Setup()
```